



Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

LAB 01

***GitHub Identity, Student Portal Registration,
Ubuntu Server Installation & SSH***

Name	Sidra-Tul-Muntaha
Registration Number	23-BSCS-084
Course Code	CS-649
Section	A
GitHub Username	sidra2244

Task 1 – GitHub Account

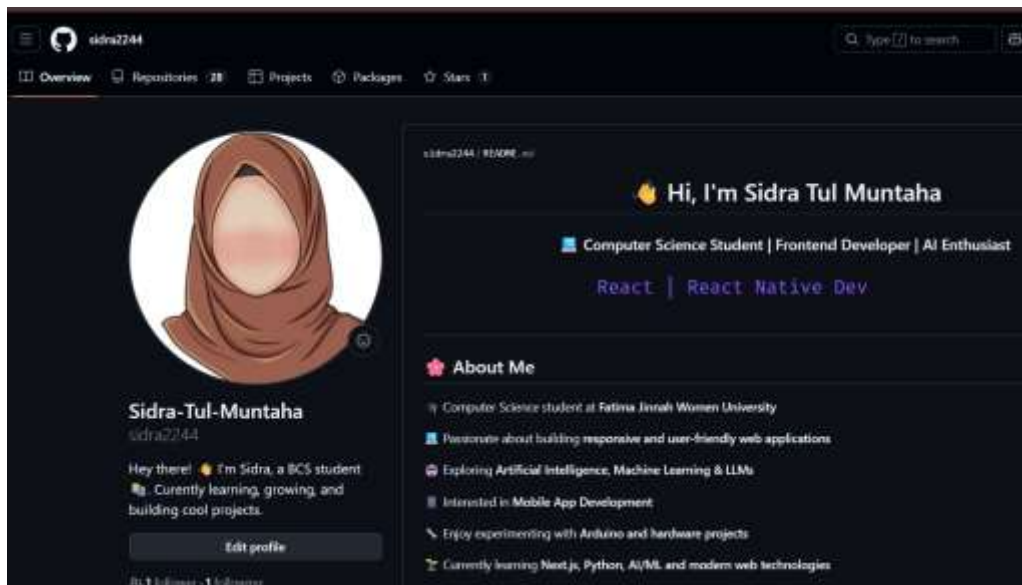


Figure: GitHub profile showing the complete actual name and GitHub username.



Figure: GitHub Settings/Public Profile showing the complete actual Name field.

Task 2 – Student Marks Portal

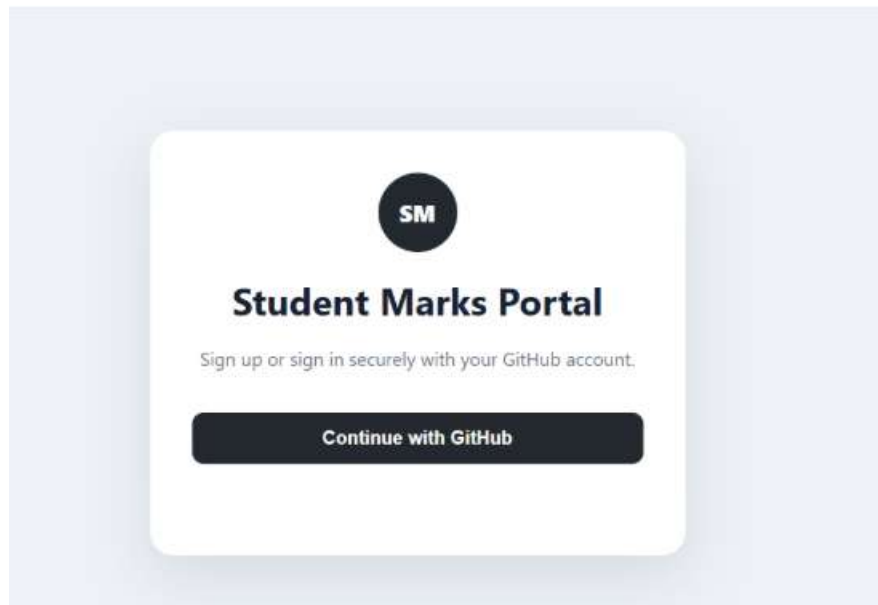


Figure: Student Marks Portal login page.



Figure: GitHub profile information shown inside the Student Marks Portal.

APPROVED COURSES			
Course Marks			
Cloud DevOps (CS-649) Section A			
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Marks have not been published.			
Total	—	—	—

Figure: Submitted enrollment status.

Task 3 – VMware & Ubuntu ISO

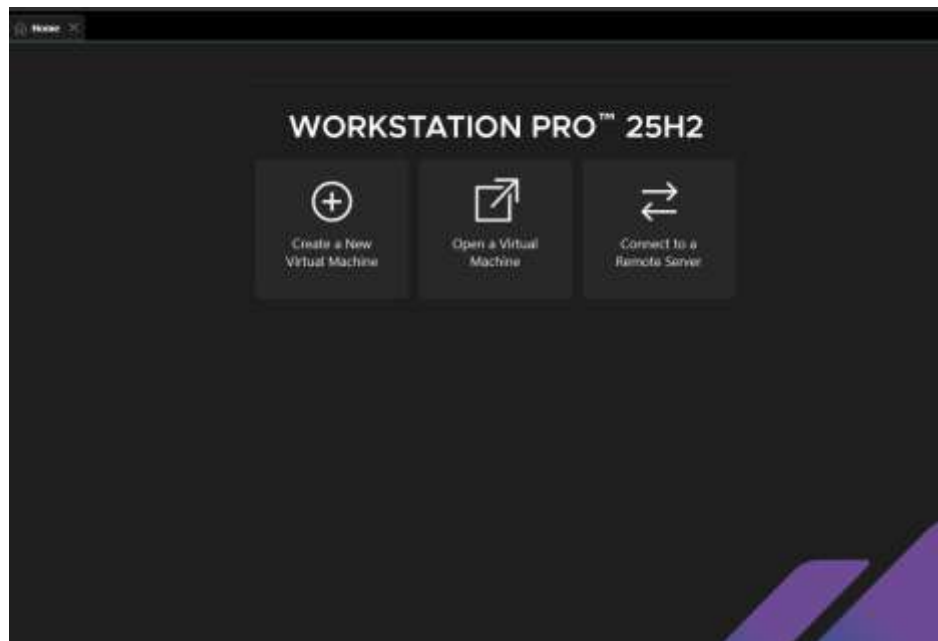


Figure: VMware Workstation Pro running.



Figure: Downloaded Ubuntu Server ISO filename.

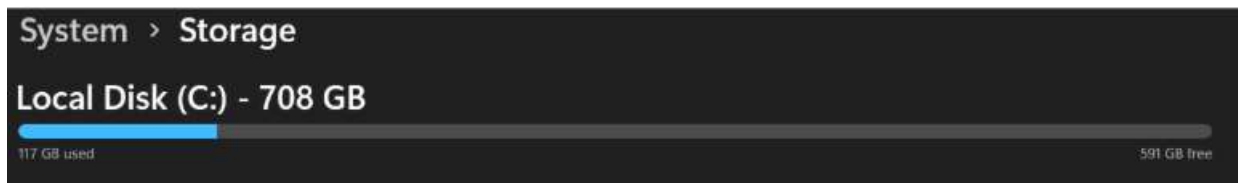


Figure: At least 20 GB free storage available.

Task 4 – Create Ubuntu VM

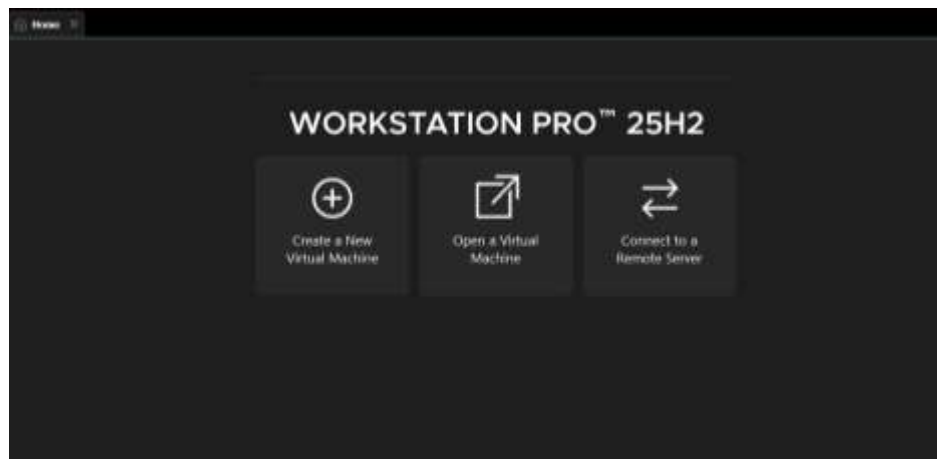


Figure: New Virtual Machine Wizard.



Figure: Typical (recommended) virtual machine configuration selected.

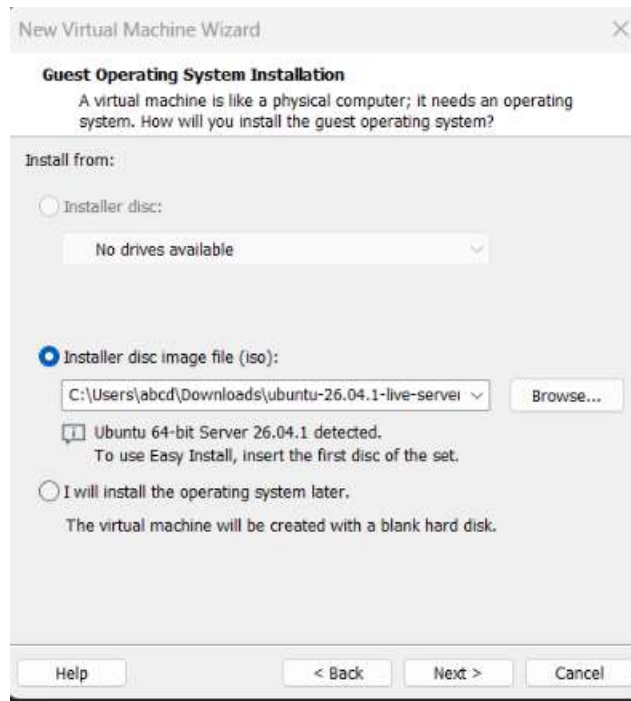


Figure: Ubuntu Server ISO selected in VMware.

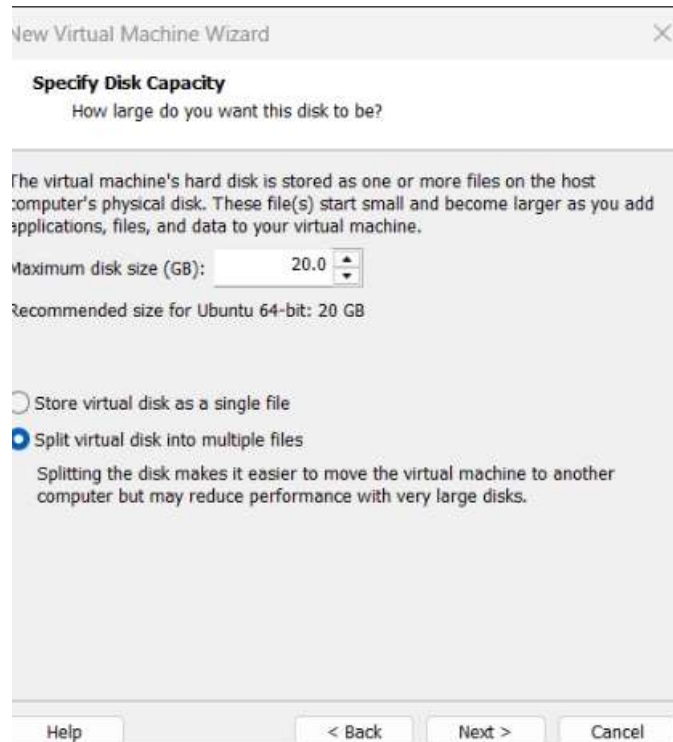


Figure: Final VM configuration with at least 20 GB virtual disk.

```
OK | Finished systemd-tmpfiles-setup-de. Create Static Device Nodes in /dev graceful
Starting systemd-sysusers.service - Create System Users...
OK | Finished systemd-random-seed.service - Load/Save OS Random Seed.
OK | Finished systemd-sysusers.service - Create System Users.
Starting systemd-resolved.service - Network Name Resolution...
OK | Finished systemd-tmpfiles-setup-de. Create Static Device Nodes in /dev
OK | Finished systemd-journal-flush.service - Flush Journal to Persistent Storage
OK | Reached target local-fs-pre.target. Preparation for Local File System
Starting systemd-udev.service - Manager for Device Events and Files
OK | Started systemd-resolved.service - Network Name Resolution.
OK | Reached target nss-lookup.target - Host and Network Name Lookups.
OK | Started systemd-udev.service - Manager for Device Events and Files
OK | Finished systemd-udev-trigger.service - Coldplug All udev Devices.
OK | Started systemd-ask-password-console.service - Requests to Console Directory Watch
OK | Reached target cryptsetup.target - Local Encrypted Volumes.
Mounting snap-core24-1643.mount - Mount unit for core24, revision 1643
Mounting snap-hwctl-123.mount - Mount unit for hwctl, revision 123.
Mounting snap-snapd-27718.mount - Mount unit for snapd, revision 27718
OK | Mounted snap-core24-1643.mount - Mount unit for core24, revision 1643
OK | Mounted snap-snapd-27718.mount - Mount unit for snapd, revision 27718
OK | Mounted snap-hwctl-123.mount - Mount unit for hwctl, revision 123.
OK | Reached target snapd.mounts.target - Mounted snaps.
OK | Started cloud-init-main.service - Cloud-init: Single Process.
```

Figure: Ubuntu Server installer boot screen.

Task 5 – Installation Language

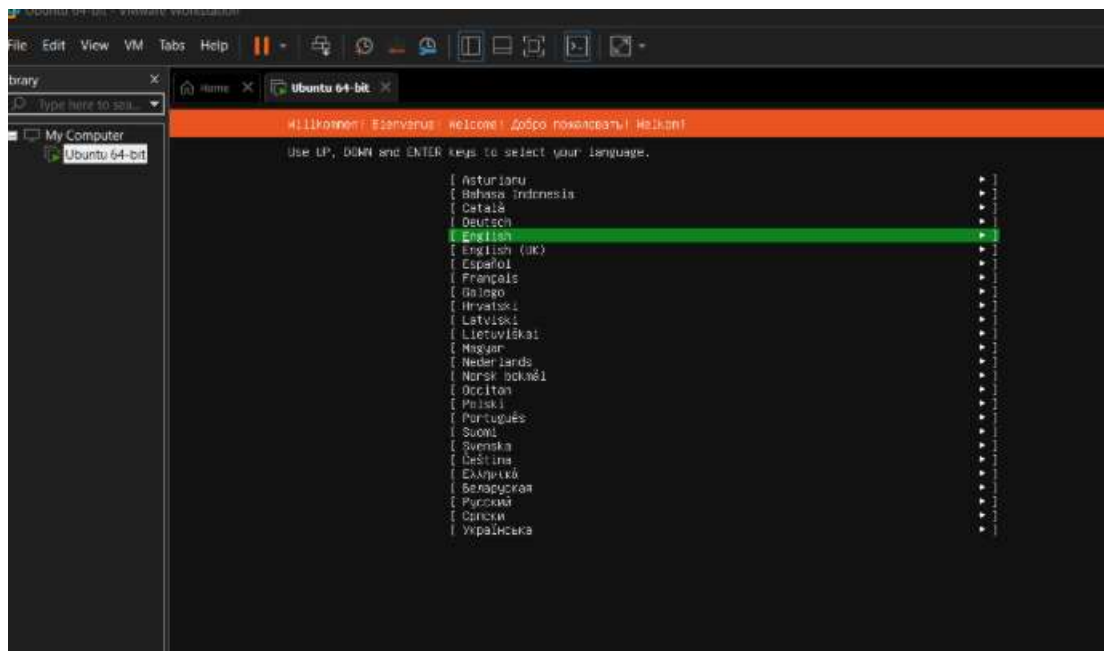


Figure: Selected Ubuntu Server installation language.

Task 6 – Keyboard

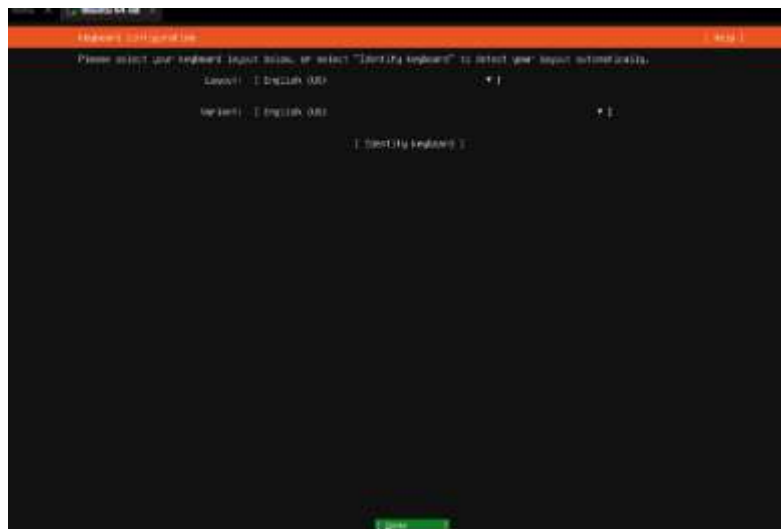


Figure: Selected keyboard layout.

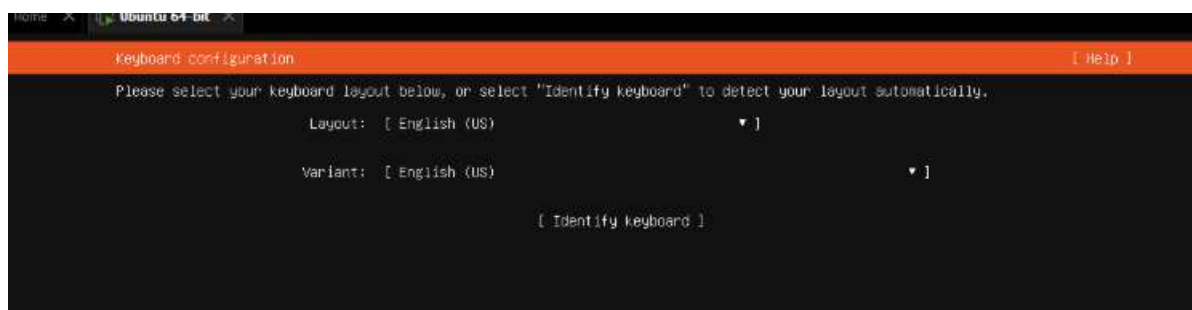


Figure: Selected keyboard variant.

Task 7 – Installation Type



Figure: Normal Ubuntu Server installation selected.

Task 8 – Network

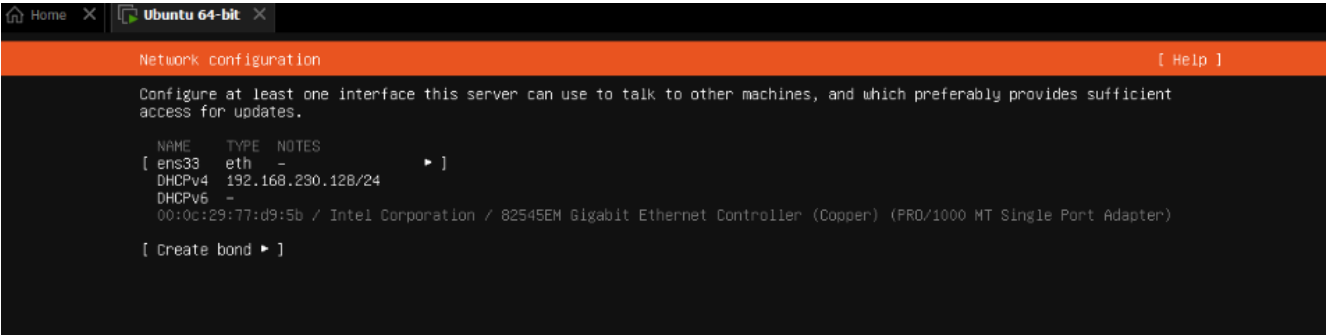


Figure: Detected network interface in the Ubuntu Server installer.

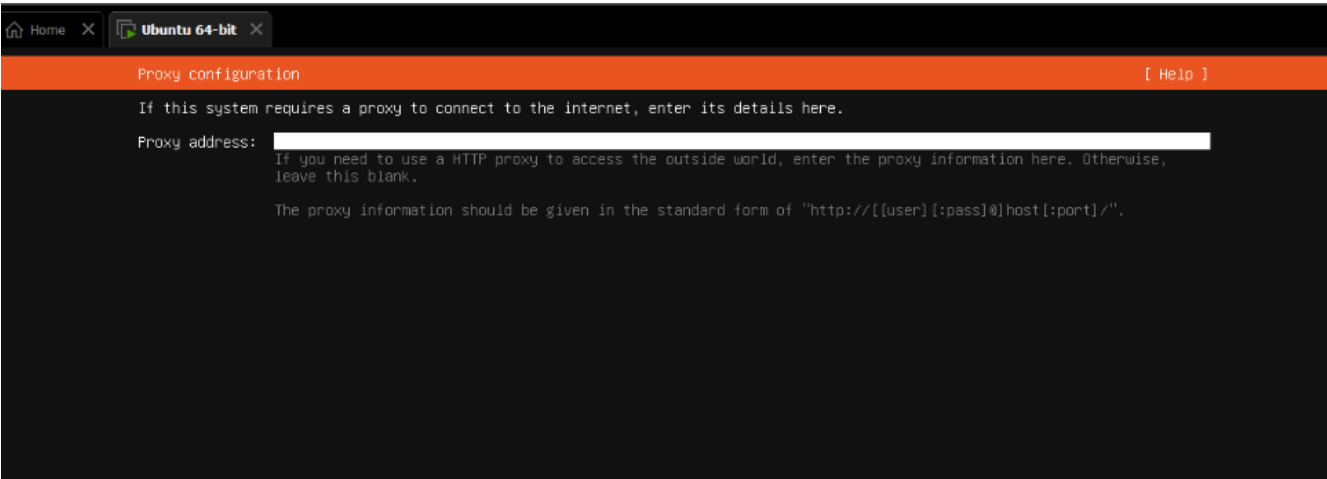


Figure: Network interface received an IPv4 address through DHCP.

Task 9 – Storage

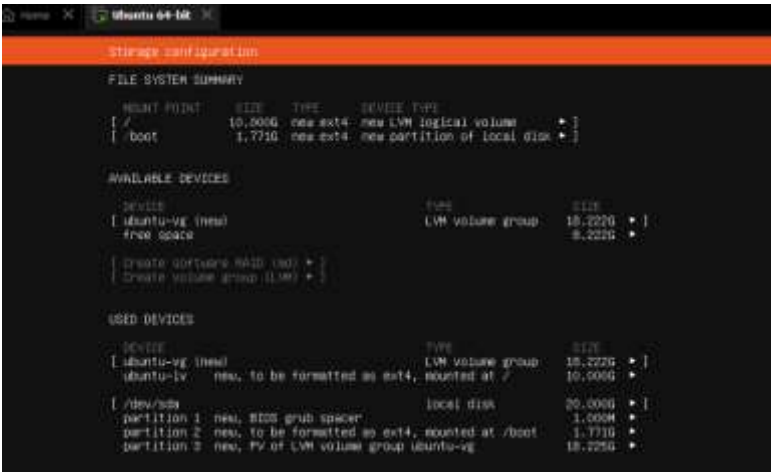


Figure: Guided storage option selected.

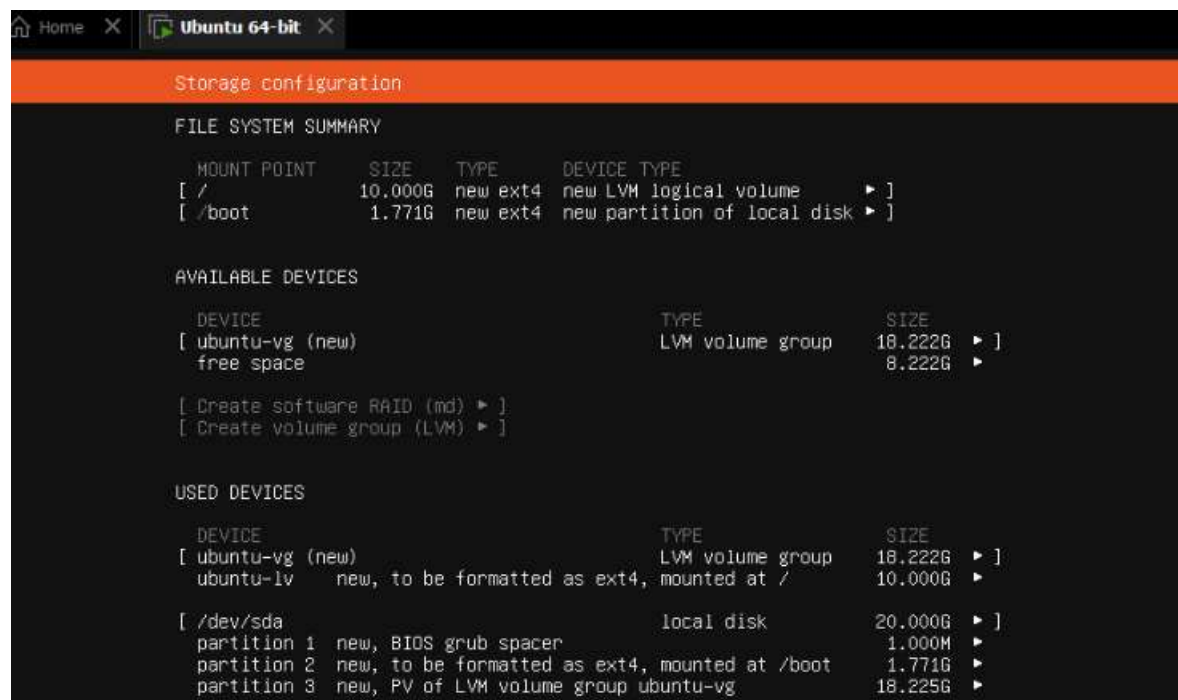


Figure: Reviewed Ubuntu Server storage configuration.

Task 10 – Select the Installation Disk

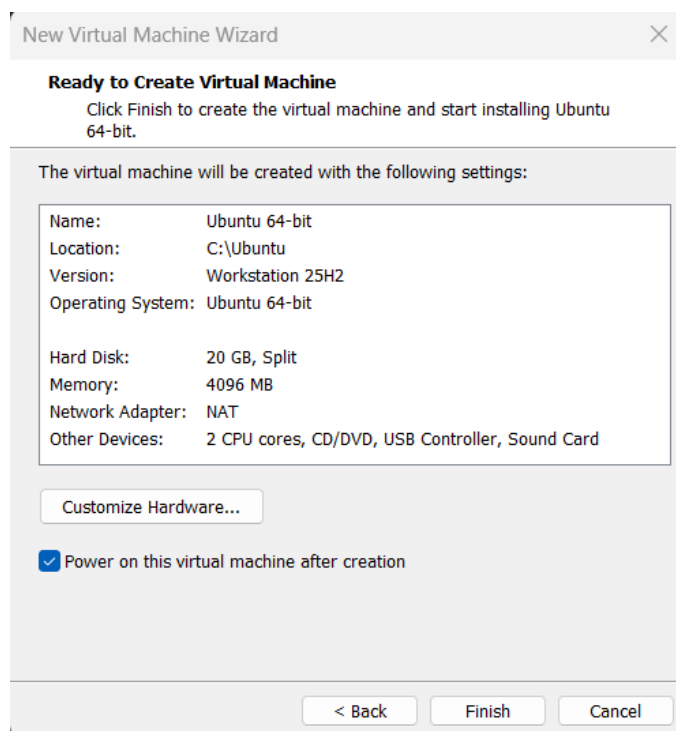


Figure: Correct Ubuntu Server installation disk selected.

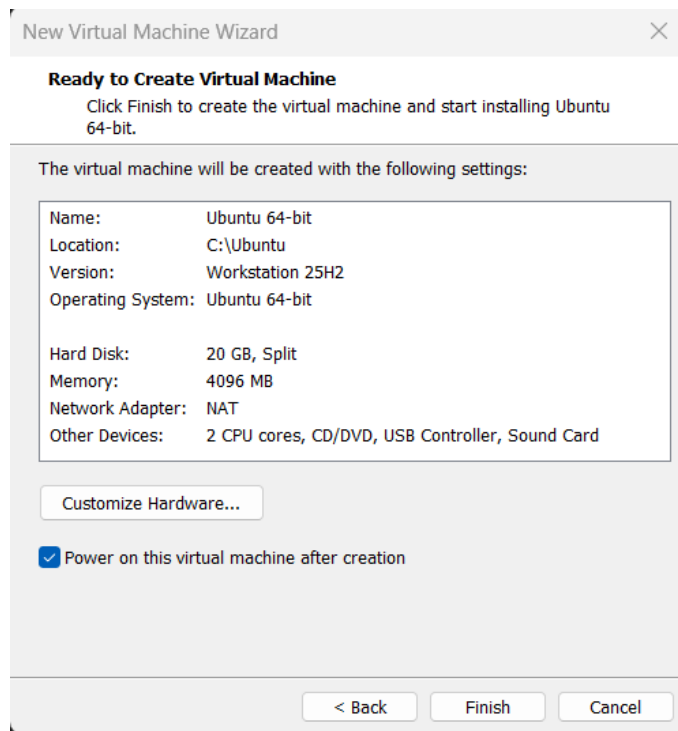


Figure: Selected disk details.

Task 11 – Review the Partitions

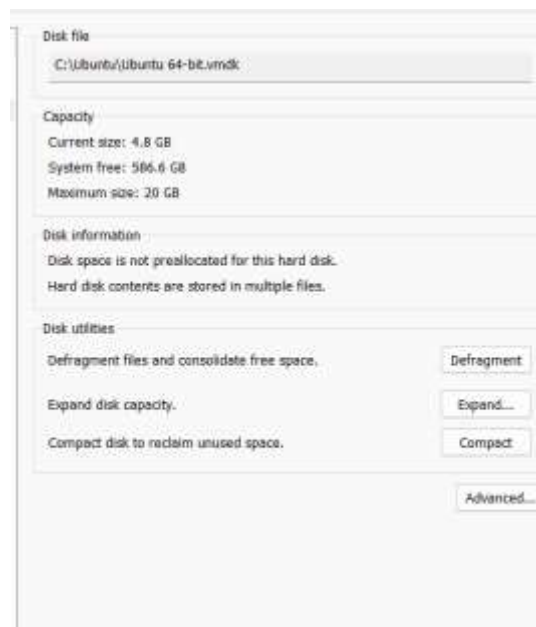


Figure: Reviewed partition layout before installation.

Task 12 – Confirm the Storage Changes

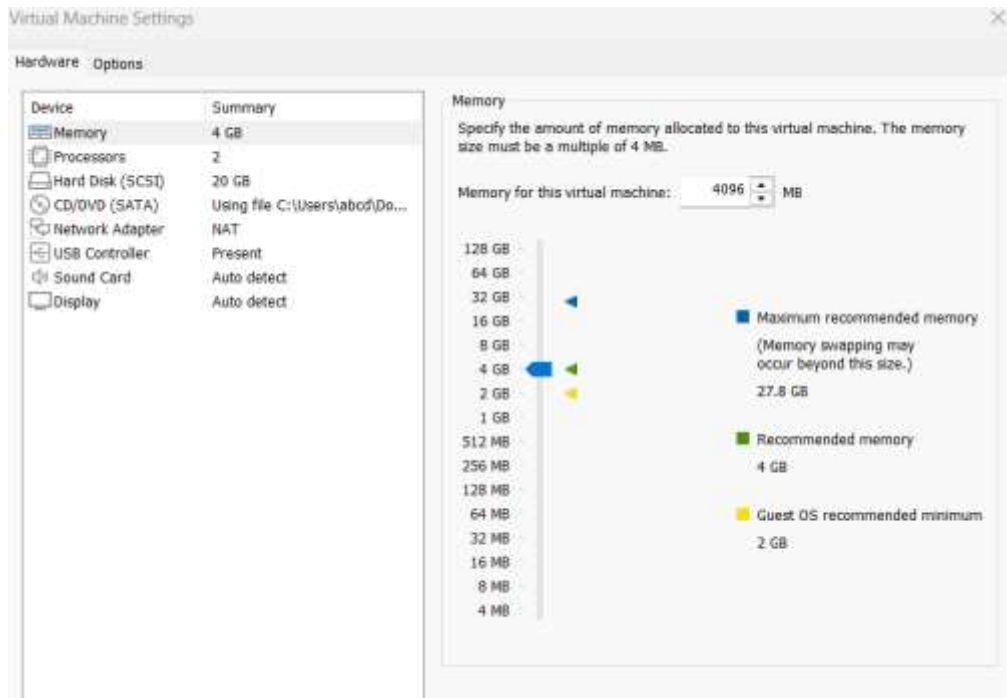


Figure: Storage changes confirmed.

Task 13 – Configure the Ubuntu Profile

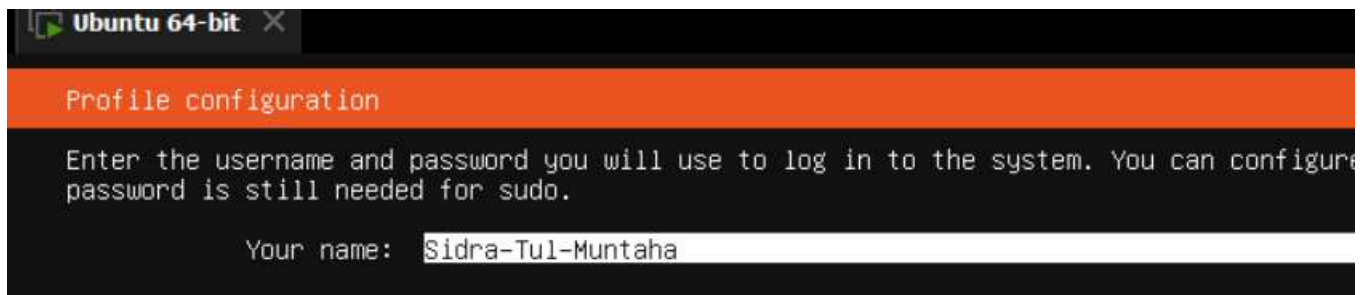


Figure: Ubuntu actual name entered.



Figure: Ubuntu server name set to ubuntu.



Figure: Ubuntu username set to sidra2244.

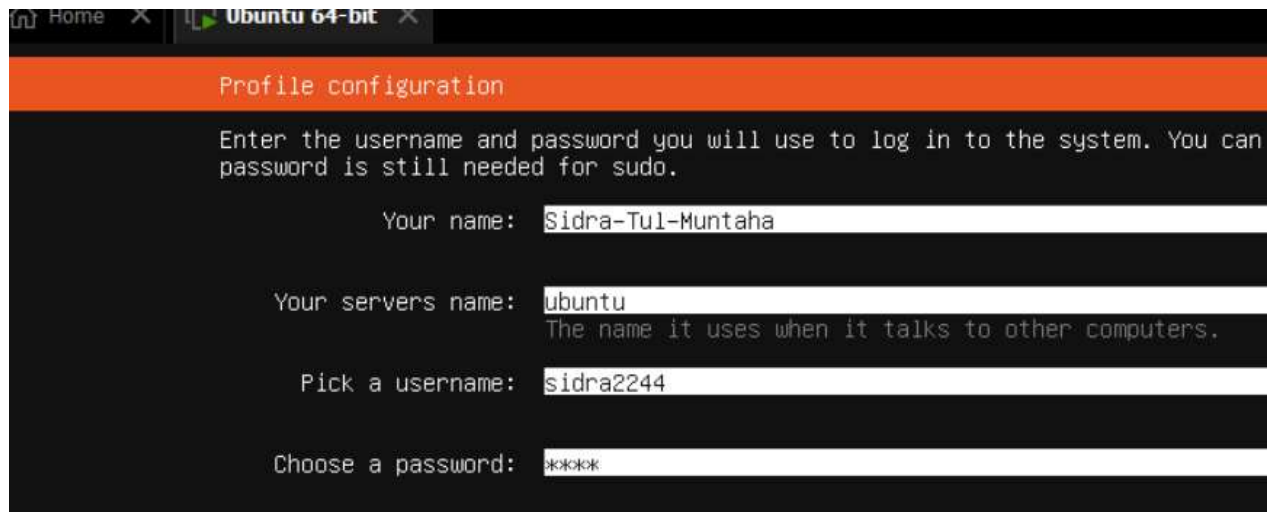


Figure: Completed Ubuntu profile setup.

Task 14 – Software Installation

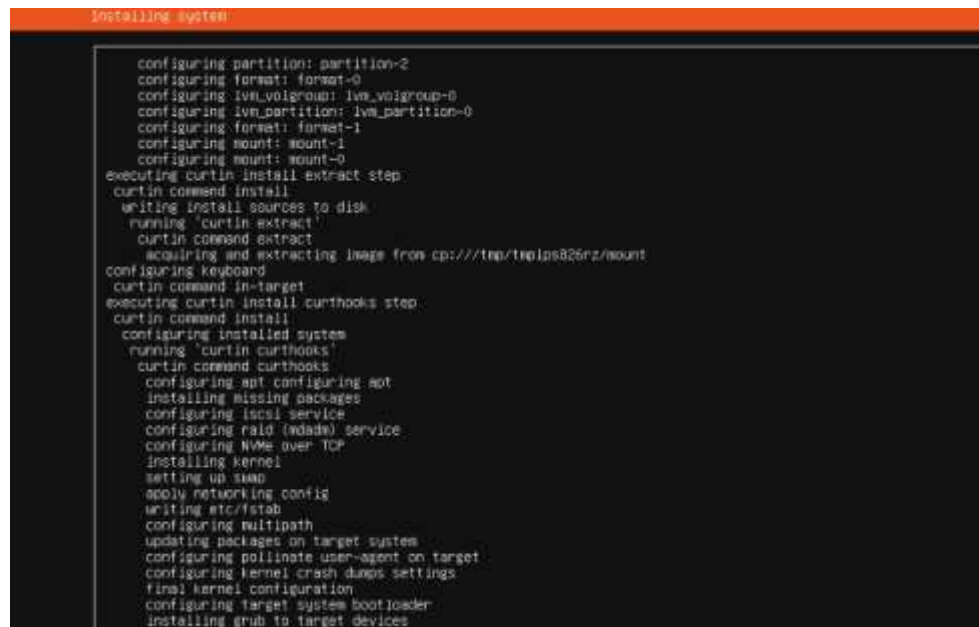
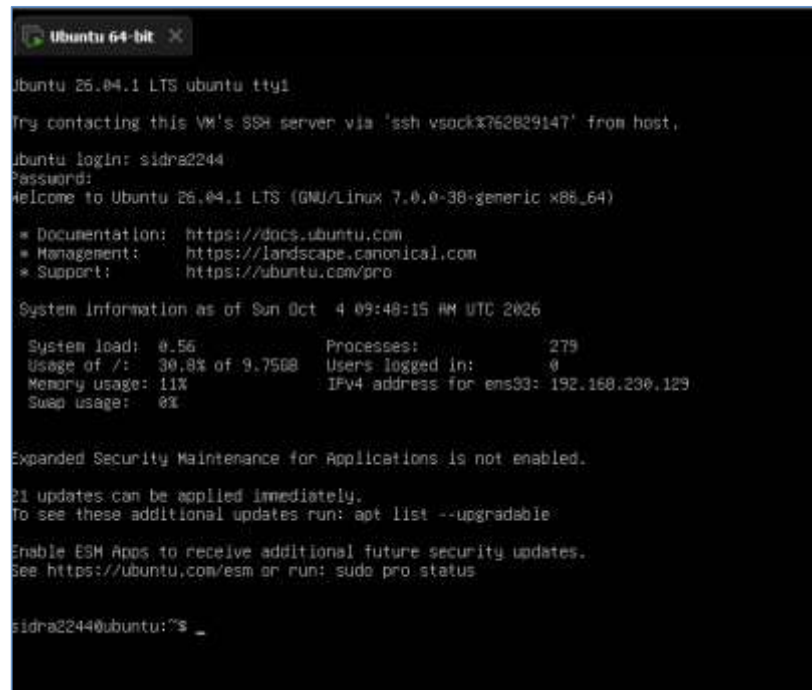


Figure: Ubuntu Server software installation in progress.

Task 15 – Reboot & Identity



```
Ubuntu 64-bit X
ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock%762829147' from host.
ubuntu login: sidra2244
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Oct  4 09:48:15 AM UTC 2026

System load:  0.56               Processes:           279
Usage of /:   30.8% of 9.75GB    Users logged in:    0
Memory usage: 11%               IPv4 address for ens33: 192.168.230.129
Swap usage:   0%

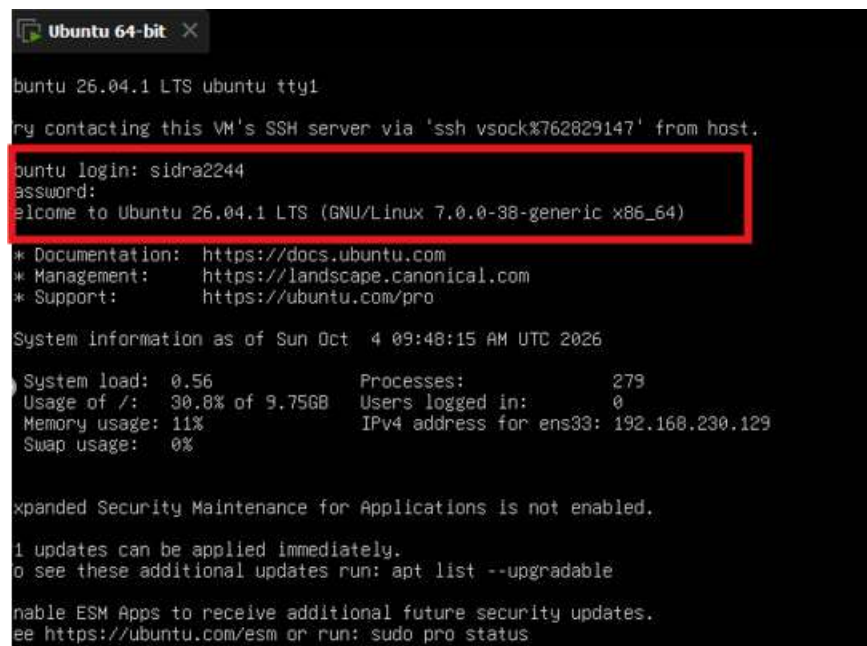
Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sidra2244@ubuntu:~$
```

Figure: Installation completed / Reboot Now screen.



```
Ubuntu 64-bit X
ubuntu 26.04.1 LTS ubuntu tty1
Try contacting this VM's SSH server via 'ssh vsock%762829147' from host.
ubuntu login: sidra2244
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Figure: Successful Ubuntu terminal login.

```

ubuntu login: sidra2244
password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System Information as of Sun Oct  4 09:48:15 AM UTC 2026

System load:  0.56               Processes:            279
Usage of /:   30.8% of 9.75GB    Users logged in:     0
Memory usage: 11%               IPv4 address for ens33: 192.168.230.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

1 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

Figure: Complete terminal prompt showing sidra2244@ubuntu:~\$.

Task 16 – IP Address

```

sidra2244@ubuntu:~$
sidra2244@ubuntu:~$
sidra2244@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

```

Figure: ip addr command and its output.

```

2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc prio_fast state UP group default qlen 1000
    link/ether 00:0c:29:77:d9:5b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx000c2977d95b
    inet 192.168.230.129/24 metric 100 brd 192.168.230.255 scope global dynamic ens33
        valid_lft 1404sec preferred_lft 1404sec
    inet6 fe80::c297:77d9:5b::/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sidra2244@ubuntu:~$ _

```

Figure: Correct Ubuntu Server IPv4 address shown after inet, with prompt visible.

Task 17 – Windows SSH

```

C:\Users\abcd>ssh sidra2244@192.168.230.129

```

Figure: Windows Command Prompt/PowerShell showing the SSH command.

```
C:\Users\abcd>ssh sidra2244@192.168.230.129
The authenticity of host '192.168.230.129 (192.168.230.129)' can't be established.
ED25519 key fingerprint is SHA256:+s3hm/T9FgGrn8q3Js+MjFONxeon7lPSt/IMhQkH6Jo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Figure: Accepted SSH server fingerprint.

```
C:\Users\abcd>ssh sidra2244@192.168.230.129
The authenticity of host '192.168.230.129 (192.168.230.129)' can't be established.
ED25519 key fingerprint is SHA256:+s3hm/T9FgGrn8q3Js+MjFONxeon7lPSt/IMhQkH6Jo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.230.129' (ED25519) to the list of known hosts.
sidra2244@192.168.230.129's password:
```

Figure: Successful SSH login without exposing the password.

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Oct  4 09:50:28 AM UTC 2026

System load:  0.06          Processes:            283
Usage of /:   30.8% of 9.75GB Users logged in:          0
Memory usage: 11%          IPv4 address for ens33: 192.168.230.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Oct  4 09:09:14 2026 from 192.168.230.1
sidra2244@ubuntu:~$
```

Figure: Remote SSH prompt showing sidra2244@ubuntu.